

```
!wget https://raw.githubusercontent.com/mwaskom/seaborn-data/master/tips.csv
```

```
busercontent.com/mwaskom/seaborn-data/master/tips.csv
```

```
ithubusercontent.com)... 185.199.108.133, 185.199.109.133, 185.199.110.133  
aw.githubusercontent.com)|185.199.108.133|:443... connected.  
0 OK
```

```
==>] 9.50K --.-KB/s in 0s
```

```
' saved [9729/9729]
```

```
from pyspark.sql import SparkSession  
  
spark = SparkSession.builder.appName("BigDataAnalysis").getOrCreate()  
  
df = spark.read.csv("tips.csv", header=True, inferSchema=True)  
  
df.show(5)  
df.printSchema()  
  
print("Total Records:", df.count())  
  
df.describe().show()
```

```
+-----+-----+-----+-----+-----+-----+-----+  
|total_bill| tip| sex|smoker|day| time|size|  
+-----+-----+-----+-----+-----+-----+-----+  
| 16.99|1.01|Female| No|Sun|Dinner| 2|  
| 10.34|1.66| Male| No|Sun|Dinner| 3|  
| 21.01| 3.5| Male| No|Sun|Dinner| 3|  
| 23.68|3.31| Male| No|Sun|Dinner| 2|  
| 24.59|3.61|Female| No|Sun|Dinner| 4|  
+-----+-----+-----+-----+-----+-----+-----+
```

only showing top 5 rows

root

```
|-- total_bill: double (nullable = true)  
|-- tip: double (nullable = true)  
|-- sex: string (nullable = true)  
|-- smoker: string (nullable = true)  
|-- day: string (nullable = true)  
|-- time: string (nullable = true)  
|-- size: integer (nullable = true)
```

Total Records: 244

```
+-----+-----+-----+-----+-----+-----+-----+  
|summary| total_bill| tip| sex|smoker| day| time|  
+-----+-----+-----+-----+-----+-----+-----+  
| count| 244| 244| 244| 244| 244| 244|  
| mean|19.785942622950824|2.9982786885245902| NULL| NULL|NULL| NULL|  
| stddev| 8.902411954856857|1.3836381890011815| NULL| NULL|NULL| NULL|
```

|     |       |      |        |     |      |        |
|-----|-------|------|--------|-----|------|--------|
| min | 3.07  | 1.0  | Female | No  | Fri  | Dinner |
| max | 50.81 | 10.0 | Male   | Yes | Thur | Lunch  |

```
: https://raw.githubusercontent.com/datasciencedojo/datasets/master/titanic.csv
```

```
--2026-06-04 06:26:05-- https://raw.githubusercontent.com/datasciencedojo/datasets/master/titanic.csv
Resolving raw.githubusercontent.com (raw.githubusercontent.com)... 185.19
Connecting to raw.githubusercontent.com (raw.githubusercontent.com)|185.1
HTTP request sent, awaiting response... 200 OK
Length: 60302 (59K) [text/plain]
Saving to: 'titanic.csv'
```

```
titanic.csv          100%[=====>]  58.89K  --.-KB/s    in 0.
```

```
2026-06-04 06:26:05 (1.93 MB/s) - 'titanic.csv' saved [60302/60302]
```

◆ Gemini

```
df = spark.read.csv("titanic.csv", header=True, inferSchema=True)
```

```
df.show(5)
```

```
df.describe().show()
```

| PassengerId | Survived | Pclass | Name                 | Sex    | Age  | SibSp | Parch |
|-------------|----------|--------|----------------------|--------|------|-------|-------|
| 1           | 0        | 3      | Braund, Mr. Owen ... | male   | 22.0 | 1     | 0     |
| 2           | 1        | 1      | Cumings, Mrs. Joh... | female | 38.0 | 1     | 0     |
| 3           | 1        | 3      | Heikkinen, Miss. ... | female | 26.0 | 0     | 0     |
| 4           | 1        | 1      | Futrelle, Mrs. Ja... | female | 35.0 | 1     | 0     |
| 5           | 0        | 3      | Allen, Mr. Willia... | male   | 35.0 | 0     | 0     |

only showing top 5 rows

| summary | PassengerId       | Survived            | Pclass             |
|---------|-------------------|---------------------|--------------------|
| count   | 891               | 891                 | 891                |
| mean    | 446.0             | 0.38383838383838383 | 2.308641975308642  |
| stddev  | 257.3538420152301 | 0.48659245426485753 | 0.8360712409770491 |
| min     | 1                 | 0                   | 1                  |
| max     | 891               | 1                   | 3                  |

```
# Total Passengers
```

```
print("Total Passengers:", df.count())
```

```
# Survivors Count
```

```
df.groupBy("Survived").count().show()
```

```
# Gender Count
df.groupBy("Sex").count().show()

# Passenger Class Count
df.groupBy("Pclass").count().show()

# Average Fare
from pyspark.sql.functions import avg

df.select(avg("Fare")).show()
```

Total Passengers: 891

```
+-----+-----+
|Survived|count|
+-----+-----+
|      1|   342|
|      0|   549|
+-----+-----+
```

```
+-----+-----+
|   Sex|count|
+-----+-----+
|female|   314|
|  male|   577|
+-----+-----+
```

```
+-----+-----+
|Pclass|count|
+-----+-----+
|      1|   216|
|      3|   491|
|      2|   184|
+-----+-----+
```

```
+-----+
|      avg(Fare)|
+-----+
|32.2042079685746|
+-----+
```

```
# Check Missing Values
```

```
from pyspark.sql.functions import col, when, count
```

```
df.select([
    count(when(col(c).isNull(), c)).alias(c)
    for c in df.columns
]).show()
```

```
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
|PassengerId|Survived|Pclass|Name|Sex|Age|SibSp|Parch|Ticket|Fare|Cabin|E
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
|          0|        0|        0|  0|  0|177|        0|        0|        0|    0|  687|
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
```

```
# Average Age by Passenger Class
```

```
from pyspark.sql.functions import avg
```

```
df.groupBy("Pclass").agg(  
    avg("Age").alias("Average_Age")  
).show()
```

```
+-----+-----+  
|Pclass|      Average_Age|  
+-----+-----+  
|      1|38.233440860215055|  
|      3| 25.14061971830986|  
|      2| 29.87763005780347|  
+-----+-----+
```